

P-192 weak-object study: singular validation and bounded CM relations

crypto research harness

2026-10-06

Contents

Status and verdict	1
Evidence labels	3
Exact object identities	3
Valid elliptic curve	3
Singular companion: a typed non-curve object	3
Experiment A — native singular-companion recovery	4
Frozen algebra and attack path	4
Fixed public synthetic fixture	4
Success, falsification, and controls	5
Result table — PENDING	5
Experiment B — exact weighted CM class-relation census	5
Frozen order and generator set	5
Exact boundary and theorem control	6
Payoff gate and result table — PENDING	7
What will count as “weak”	7
Confidence model	7
Deterministic claims	7
Timing and implementation variability	8
Generalizing beyond one fixture	8
Prior evidence and non-duplication	8
Open obligations	8
Sources and provenance	8
Repository and protocol sources	8
External parameter and terminology sources	9

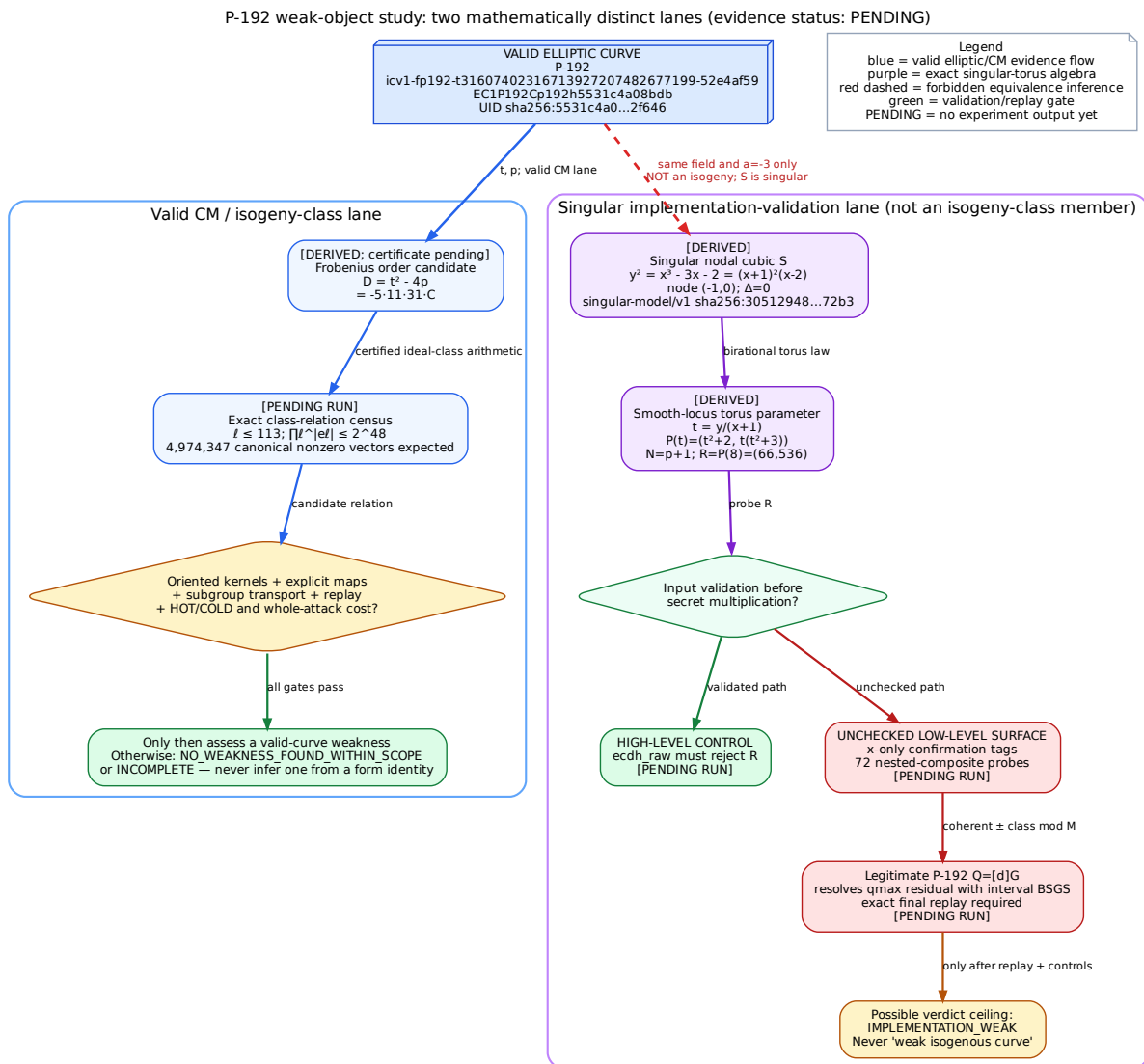
Status and verdict

Evidence cutoff: 2026-10-06. Experimental verdict: PENDING. This is the source-linked report scaffold for two approved native experiments. It contains derived algebra, frozen inputs, claim gates, and prior-art provenance. It does **not** contain an observed recovery, a completed CM census, a timing result, or a new weakness claim. Every empirical result cell remains PENDING until it is filled from hashed run artifacts and independently replayed.

The work uses one fixed, public, synthetic scalar. It does not target deployed keys, live services, or private data. Its purpose is to distinguish an implementation-validation failure from a property of a valid elliptic curve.

Two hypotheses are deliberately kept separate:

1. **Singular implementation-validation lane.** A nodal cubic shares the P-192 field and coefficient $a=-3$, but it is not an elliptic curve and is not isogenous to P-192. The experiment asks whether an unchecked low-level multiplication surface can turn x-only confirmation tags into recovery of the fixed synthetic scalar. The strongest admissible verdict is IMPLEMENTATION_WEAK.
2. **Valid CM/isogeny-class lane.** An exact, bounded class-relation census is run in the endomorphism order attached to the valid P-192 isogeny class. A form relation is not a weakness: global orientation, explicit maps, subgroup transport, point replay, and fully charged payoff must all pass. The preregistered negative verdict is only NO_WEAKNESS_FOUND_WITHIN_SCOPE for the frozen box.



Evidence labels

Statements in this report use the following labels.

label	meaning
FROZEN	Fixed by approved protocol before execution.
DERIVED	Algebraically derived from stated inputs; not a run measurement.
INDEPENDENTLY VERIFIED — PRIOR	Replayed evidence from an earlier committed/archived run.
MEASURED	Hashed native run plus replay receipt. No new item has this label yet.
PENDING	Required evidence does not yet exist in this report package.

Exact object identities

Valid elliptic curve

The valid object is P-192, registered as `icv1-fp192-t31607402316713927207482677199-52e4af59`.

- **Field prime p (FROZEN):**

6277101735386680763835789423207666416083908700390324961279.

- **Short-Weierstrass coefficients (FROZEN):** $a=p-3$; exact b :

$b=2455155546008943817740293915197451784769108058161191238065$.

- **Prime group order n (FROZEN):**

6277101735386680763835789423176059013767194773182842284081.

- **Trace $t=p+1-n$ (DERIVED):** 31607402316713927207482677199.

- **Compact representation:** EC1P192Cp192h5531c4a08bdb.

- **Full representation UID:** prefix `urn:ec-record:1:sha256:` followed by
5531c4a08bdb64b6e86a6e30e9a08aa57edef7af15ac5f6d4d2a83a53bf2f646.

The exact registry source is [docs/curves/registry.json](#) at base revision:

0dc6b7f4a03253e7c7641f99700f7f423f8d07e0.

Its source SHA-256 is recorded in the artifact index.

Singular companion: a typed non-curve object

DERIVED. Define over the same field

$$S : y^2 = x^3 - 3x - 2 = (x + 1)^2(x - 2).$$

For a short-Weierstrass model, the discriminant is $-16(4a^3+27b^2)$. At $a=-3, b=-2$, the inner term is $4(-27)+27(4)=0$; therefore S is singular, with node $(-1, 0)$. It is not assigned ICV1, EC1, or a curve UID. The exact non-curve preimage is stored in [singular-model.canonical.json](#) with type `singular-model/v1`; its SHA-256 is `305129485b8a281f5cd73a7a0ec4d82a7f84f432847c1b9d5eb1734983db72b3`.

The probe $R=(66, 536)$ satisfies $536^2=66^3-3*66-2=287296$, is away from the node, and is not on the valid P-192 model because the valid b differs. Sharing p and a does not define an isogeny, an isomorphism, or a curve-family member.

Experiment A — native singular-companion recovery

Protocol: EXP-SCURVE-29040c, approved by DEC-20261006-af8214. Exact inspected protocol SHA-256: 5de49bcf72cc676290982314046c6579b10473372aed822c6955813660e9030e. Its archive/commit receipt is PENDING; see [ARTIFACT_INDEX.md](#).

Frozen algebra and attack path

On the smooth locus set $t=y/(x+1)$. The inverse parameterization and group law used by the independent arithmetic control are

$$P(t) = (t^2 + 2, t(t^2 + 3)), \quad t_1 \star t_2 = \frac{t_1 t_2 - 3}{t_1 + t_2}.$$

The identity is $t=\infty$, inverse is $t \rightarrow -t$, and $R=P(8)=(66,536)$. Because $p \bmod 12 = 11$ and -3 is a nonsquare, this is the nonsplit nodal torus with order

$$N = p + 1 = 2^{64} \cdot 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 274177 \cdot 6700417 \cdot 67280421310721.$$

The frozen point-order certificate must prove $[N]R=0$ and $[N/q]R \neq 0$ for every distinct prime factor q . It may not be inferred from the factorization alone.

The largest factor and its cofactor are

$$\begin{aligned} q_{\max} &= 67280421310721 & (\log_2 \approx 45.93525) \\ M &= 93297598515282145091939986791922449178951680 \end{aligned}$$

The x -only oracle is sign-invariant. Nested composite-order probes therefore retain one coherent pair $\{r, -r \bmod m\}$ rather than independently choosing a sign for every factor. The frozen sequence contains 64 lifts by two and eight odd factors, so the expected primary-oracle count is exactly 72. Candidate comparisons are counted separately and must not be relabelled as oracle calls.

After recovery modulo M , the legitimate public key $Q=[d]G$ on P -192 is required to solve the remaining interval. One shared negation-BSGS table is frozen at $\text{table_bits}=24$, $m=5,800,019$, stride $11,600,037$, no more than $5,800,018$ finite baby entries, and no more than $5,800,018$ giant positions per orientation. Every fingerprint hit requires exact replay.

Fixed public synthetic fixture

- **Seed:** P192-SINGULAR-V1; hex 503139322d53494e47554c41522d5631.
- **Scalar mapping:** $1 + 0S2IP_BE(\text{SHA256}(\text{domain} \parallel \text{seed})) \bmod (n-1)$.
- **Expected d in hex:**
8b70f85429c81223e88be1a60590627a3c8e558a24dd5180.
- **Expected d in decimal:**
3419090462492416794041408622244871729251768983841006244224.
- **Unordered residues modulo M :**
 - 78222222987683236539393919745863586265518464.
 - 15075375527598908552546067046058862913433216.
- **True residual k :** 36647143301682. Check only after recovery; never use it to prune the search.

Success, falsification, and controls

An IMPLEMENTATION_WEAK result requires all of the following, not merely successful torus arithmetic:

- the exact P-192 and singular-point/order certificates pass;
- the actual unchecked scalar_mul_secret surface is used for every primary query and exposes only the frozen x-derived HMAC tag;
- one coherent plus/minus class is recovered after exactly 72 primary calls;
- the legitimate P-192 residual solve returns the fixed d;
- independent final replay proves [d]G=Q;
- the safe high-level ecdh_raw path rejects every singular probe before multiplication; and
- exhaustive p=59, wrong-parameter, mutated-transcript, full-point positive, and independent-torus controls all give their preregistered verdicts.

If instrument controls pass but the x-only transcript cannot recover the coherent class, or neither residual orientation replays to Q, the hypothesis is falsified. A timeout, OOM, or missing receipt is INCOMPLETE, not evidence that the valid curve is strong.

Result table — PENDING

No speedup is asserted. Counts below are the frozen boundaries; observed values must be copied from recovery.json and phase-metrics.json only.

phase	frozen boundary/reference	observed
Valid P-192 reference	approximately sqrt(n) group work; matched native reference not run	PENDING
Nested residue recovery	exactly 72 oracle queries; comparisons separate	PENDING
Two-orientation BSGS	at most 5,800,018 babies and 5,800,018 giants/orientation	PENDING
Whole pipeline	certification, probes, comparisons, table, giants, replays	PENDING

Correctness and the ratio to a matched reference are both PENDING. A ratio is not meaningful until every phase is expressed in the same counted unit.

Wall time is secondary engineering evidence. A performance claim additionally requires the repository's isolated A/A and interleaved paired-measurement protocol; a single recovery run is a feasibility result, not a confidence interval.

Experiment B — exact weighted CM class-relation census

Protocol: EXP-SCURVE-647ade, approved by DEC-20261006-af8214. Exact inspected protocol SHA-256: 0fe122a0c6be3e2118b4d9c276a1085285c7e6a2cee38d15bbeed8099fa17145. Its archive/commit receipt is PENDING.

Frozen order and generator set

DERIVED; certificate PENDING. The Frobenius discriminant is

$$D = t^2 - 4p = -24109379060336110122544161233113975664949272517896865359515 = -5 \cdot 11 \cdot 31 \cdot C,$$

where $C=14140398275856956083603613626459809774163796198179979683$. The run must emit and replay a recursive Pocklington certificate for C. Only after that proof may it conclude that D is squarefree and fundamental, the Frobenius conductor is one, and $\text{End}(E)=\mathbb{Z}[\pi]=0_D$.

The frozen ramified roots are 5: [2], 11: [3], and 31: [7]. The frozen split roots are:

ell	roots of $X^2 - tX + p \bmod \text{ell}$
13	2, 5
23	21, 22
37	12, 35
43	8, 26
73	60, 67
89	6, 83
101	17, 70
103	5, 36
107	56, 68
113	26, 42

Each root and orientation must be recomputed. A positive split generator uses the smaller root; global conjugates are canonicalized by requiring the first nonzero split exponent to be positive. Ramified exponents are unsigned.

Exact boundary and theorem control

The census includes every canonical exponent vector with $L = \text{product } \text{ell}^{|\mathbf{e}_{\text{ell}}|} \leq 2^{48}$; empirical weights may rank but may not prune. The preregistered cardinalities are 9,948,061 oriented vectors, 4,974,348 canonical vectors including zero, and 4,974,347 canonical nonzero vectors, under a hard cap of 5,000,000 canonical states. A differing count invalidates the run.

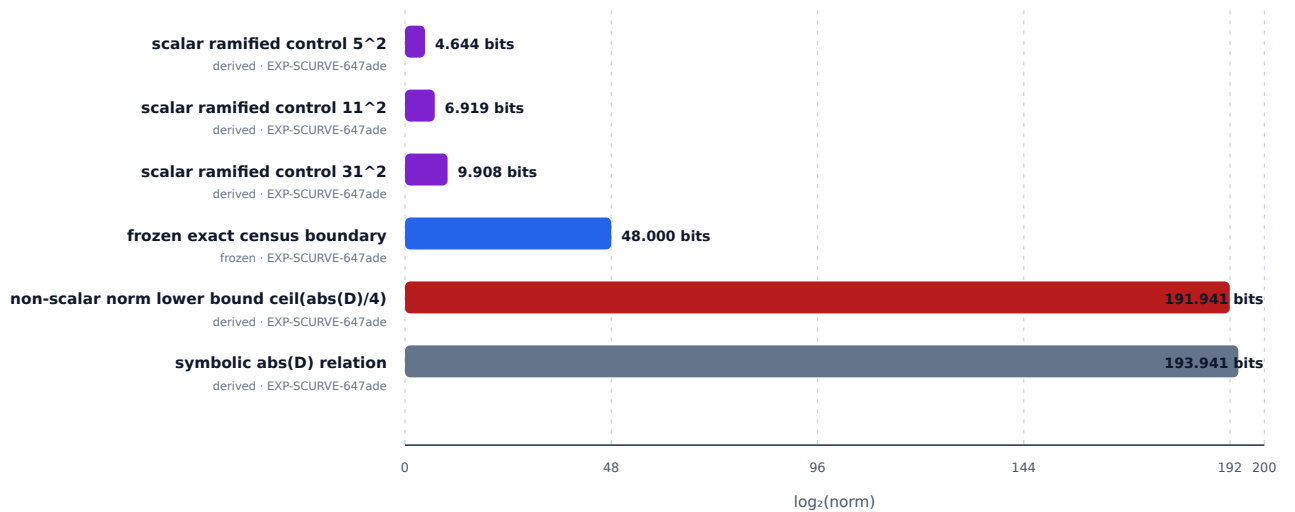
For a non-scalar $\alpha = u + v\pi$, the protocol must independently certify

$$N(\alpha) \geq \left\lceil \frac{|D|}{4} \right\rceil = 6027344765084027530636040308278493916237318129474216339879 \approx 2^{191.941}.$$

If all premises pass, the exact 2^{48} lane can contain only scalar ramified relations. This is a proof-controlled scope statement, not a claim about all relations or all attack mechanisms.

CM degree frontier: exact scope versus non-scalar theorem bound

Unit: $\log_2$ ideal norm / geometric degree. Derived and preregistered values only; no run or timing result is plotted.



The controls are g_5^2 , g_{11}^2 , and g_{31}^2 on P-192; the positive non-scalar toy relation $g_2^3 = 1$ at $D = -23$; and the symbolic P-192 relation $g_5 * g_{11} * g_{31} * g_C = 1$ with $\alpha = \sqrt{D} = 2\pi - t$. The last relation is genuine algebraically but is outside the buildable $\text{ell} \leq 113$ set: a C-degree edge is unavailable, and its action on $E(F_p)[n]$ does not by itself improve a DLP. It must remain a rejected symbolic control.

Payoff gate and result table — PENDING

Every gate is currently **PENDING**:

1. **Order/CM proof:** P-192 tuple/order, C certificate, fundamental D, and conductor one. Without it there is no search claim.
2. **Exact membership:** both enumeration orders, exact counts, and boundary digest. This can establish only finite-scope completeness.
3. **Algebraic relation replay:** reduced forms, vectors, α , norm, lambda, and ideal-HNF checks. This can establish only an algebraic identity.
4. **Global map realization:** oriented kernels, explicit chains and endpoint IDs, maps, and point replay. This establishes buildability, not payoff.
5. **HOT/COLD accounting:** raw operation tuples and calibration dispersion, including construction, evaluation, verification, and serialization.
6. **Whole payoff:** a fully charged canonicalization or DLP inequality against the declared baseline. Only this gate can support a valid-curve weakness.

The current native implementation may explicitly report map calibration as unsupported. If so, the exact census can still establish a bounded negative algebraic result, but the map and payoff layers remain INCOMPLETE; no proxy cost may be substituted for measured HOT/COLD tuples.

What will count as “weak”

The word *weak* is reserved for an auditable statement with an exact object, threat model, boundary, and verified advantage.

- **REJECTED_AS_DEIGNED:** the validated public path rejects the singular probe before secret multiplication. This says nothing about unchecked low-level callers.
- **IMPLEMENTATION_WEAK:** the actual unchecked surface, fixed x-only oracle, end-to-end planted-key recovery, independent replay, and every control pass. The singular cubic is still not a weak elliptic or isogenous curve.
- **STRUCTURAL_CURVE_WEAK:** a valid nonsingular curve identity, certified map/attack, subgroup transport, verified solve, complete operation accounting, and material advantage over a matched reference all pass. A small coefficient, form relation, or walk alone is insufficient.
- **NO_WEAKNESS_FOUND_WITHIN_SCOPE:** the frozen CM box is exhausted with exact counts, proofs, and replay, and no candidate passes the buildability and payoff gates. This is not a proof of global security or absence beyond the box.
- **INCOMPLETE:** any mandatory certificate, map, cost, control, or replay is missing. Resource exhaustion is not strength evidence.

P-192's roughly 96-bit generic security level and legacy status are baseline properties, not discoveries of this campaign.

Confidence model

Deterministic claims

Primality, factorization use, point order, class-group products, HNF identities, enumeration membership, map equations, and final scalar replay are deterministic claims. Statistical confidence is the wrong tool for them. Confidence comes from self-contained certificates, exact counts/digests, a second enumeration order, mutation-negative controls, and an independent verifier that does not trust the producer's intermediate state.

An exhaustive negative statement is therefore conditional but crisp: *given the certified order and the exact generator/boundary definition, every state in that finite set was enumerated and no state passed the declared gate*. It cannot be widened by a confidence interval.

Timing and implementation variability

Any wall-clock comparison must first measure A/A spread, then interleave baseline and candidate for at least five paired rounds under the repository's isolated runner. Report counted native operations as primary, plus median, minimum, contention flags, and a 95% paired confidence interval. A runtime improvement is supported only if the interval excludes no improvement and all outputs/digests match. No such comparison is present yet.

Generalizing beyond one fixture

One fixed synthetic scalar can demonstrate a mechanism and exact recovery; it cannot estimate a prevalence rate across keys, implementations, or curves. A later population claim would require a preregistered sampling frame, independent holdouts, multiple keys per implementation, explicit failure retention, exact binomial intervals, and multiplicity control across curve/candidate searches. No such population inference is made here.

Prior evidence and non-duplication

The prior committed P-192 isogeny walk [research/isogeny_walk_p192_20261004/README.md](https://research.isogeny_walk_p192_20261004/README.md) is **INDEPENDENTLY VERIFIED — PRIOR**, not a result of these experiments. Its 20,000-curve run is p192-71c04205135fcf8f, archived under s3://crypto-autoresearcher/isogeny-walk/runs/p192-71c04205135fcf8f. It reported 20,000 valid curve records, 128,700 edges, zero failures, and replay pass, but explicitly established no DLP weakness or per-curve IC difference. The new exact CM census asks a different, relation-focused question and must not rewrite that frozen evidence.

The repository's typed-link rules (docs/curves/ic/curve-links/README.md; source hash in the artifact index) require exact endpoints, kernels/maps, dual status, subgroup transport, and replay before an isogeny route can carry a DLP claim. That rule is the basis of the CM payoff gate and the red “NOT an isogeny” separation in the diagram.

Open obligations

- Execute and independently replay EXP-SCURVE-29040c at the committed native revision; fill the result table from immutable artifacts.
- Execute the full exact census for EXP-SCURVE-647ade; preserve the second enumeration-order digest and every excluded relation class.
- Treat explicit global maps and HOT/COLD calibration as open if the backend reports them unsupported. Do not fabricate them from the degree-linear proxy.
- Regenerate this report and PDF after evidence lands, then hash and inspect all derived artifacts as described in the artifact index.
- Do not extend the conclusion to sect113r1, other standardized curves, other APIs, or a population of implementations without separately frozen inputs and evidence.

Sources and provenance

Repository and protocol sources

1. P-192 registry entry: docs/curves/registry.json, exact EC1/UID above; source hash and revision recorded in the artifact index.
2. Repository curve parameters: src/ecc/curve_zoo.rs, source hash recorded in the artifact index.
3. Prior P-192 walk: research/isogeny_walk_p192_20261004/README.md, with report, run, and store hashes in the artifact index.
4. Frozen experiment records EXP-SCURVE-29040c and EXP-SCURVE-647ade, approved by DEC-20261006-af8214; exact inspected hashes are recorded above.
5. Evidence and accounting rules: AGENTS.md, SHA-256 recorded in the artifact index.

External parameter and terminology sources

- NIST, [FIPS 186-4](#), archived standard containing the legacy NIST prime-curve parameters.
- SECG, [SEC 2 version 2.0](#), secp192r1 domain parameters.
- Andrew V. Sutherland, "[Isogeny Volcanoes](#)", for isogeny-volcano terminology. The report does not treat terminology as a certificate for any new edge.

The complete expected artifact set and all PENDING receipts are in [ARTIFACT_INDEX.md](#).